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POSTER

From Simulation to Reality: China's First Authentic-Scenario Emergency Drill for Preserving Licensed Electronic Journals in Biomedicine

A 25-Library Collaboration to Validate the National Digital Preservation Infrastructure under Real-World Conditions

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1 Background & Challenges

Digital preservation systems often function as "dark archives," where data retrieval capabilities under real-world loads remain untested. In the biomedical sector, reliance on commercial publisher platforms creates vulnerability to cyberattacks or geopolitical disruptions. To address this, the National Digital Preservation Center (NDPC) Medical Sub-Centre launched the world's first multi-institutional, authentic-scenario emergency drill involving real users, 25 libraries, and a commercial publisher (Karger).

The Paradox of "*Dark Archives*": The majority of preservation systems function as "dark archives," wherein data is securely stored, but the ability to effectively retrieve and utilize this data under real-world conditions—encompassing actual users, network environments, and system loads—remains largely untested.

Risks of Publisher Dependency: Access to high-quality commercial electronic resources relies heavily on publisher platforms. These platforms are vulnerable to unpredictable disruptions—including subscription discontinuation, platform crashes, cyberattacks, or geopolitical events. Such disruptions threaten continuous access to critical biomedical literature.

Validation Deficiency: Emergency preparedness drills conducted under realistic scenarios are uncommon. Prior to this initiative, no multi-institutional emergency drill involving real users had been implemented in China's domestic biomedical sector. This drill serves to fill that gap, validating the newly established NDPC Medical Sub-Center (inaugurated Oct 30, 2024).

2 A Milestone Emergency Drill

To address these challenges, the NDPC Medical Branch launched the world's first multi-institutional, authentic-scenario emergency drill specifically designed for licensed biomedical electronic journals.

10+	25	8h
Years Since NDPP Launch	Participating Libraries	Duration

Objectives:

1) Validate Feasibility: Test emergency access protocols in authentic operational environments; **2) Evaluate Performance:** Assess concurrency, retrieval, and download speeds under real user load; **3) Identify Vulnerabilities:** Expose weaknesses in workflows and technical infrastructure.

3 Design & Execution

Drill Method: The team established working groups and coordinated IP authorization transfers. A key challenge was ensuring that the "emergency switch" could be toggled without violating licensing agreements. This required establishing a trusted framework between the NSTL (National Center), the libraries (Users), and Karger (Content Provider)

1 Preparation Phase
Establishment of working groups, coordination of resources, and delivery of technical training on NDPC system architecture

2 Pre-Drill Testing
Conducted stress tests, access rights validation, and troubleshooting to ensure operational readiness

3 Official Drill Execution
Executed an 8-hour comprehensive live drill simulating access disruption to Karger publisher resources. Simultaneous activation of NDPC emergency access by 25 participating libraries. Included manual testing and continuous system monitoring.

4 Post-Drill Evaluation
Systematic collection of feedback and thorough assessment of system performance, user experience, and infrastructure resilience. Preparation of detailed reports and implementation of service optimizations.

Rationale for Selecting Karger Publishers: First, Karger's biomedical portfolio closely aligns with the mission of the NDPC Medical Branch. Second, its licensing agreements included clearly defined archival preservation rights, making it legally feasible to conduct an emergency access activation without contractual ambiguity. Third, Karger expressed strong institutional support for validating long-term preservation readiness

4 Key Findings & Improvements

- Karger Electronic Journals
- Full-text Search & Browse
- PDF Download Services
- Data Security Validation

100%	25/25
System Availability	Libraries Activated

System Performance

During the 8-hour window, the system successfully supported full-text search, browsing, and PDF download services, across all 25 activated libraries.

Technical Insights

A significant technical value of "real-world" testing was demonstrated during the pre-drill phase. A "cache overflow" issue was identified, where high-concurrency PDF downloads caused failures (0kb files). This was resolved by implementing a scheduled cache-clearing script—a solution that likely would not have been discovered in a static "dark archive" simulation.

User Feedback

Qualitative feedback confirmed that the emergency service processes were orderly, efficient, and smooth. The cross-institutional collaboration exhibited strong coordination capabilities, and the system response was rapid, providing an excellent user experience.

5 Connection to iPRES 2026 Themes

P Preservation
This milestone validates the NDPC infrastructure's readiness by moving preservation practice from simulation to real-world application, demonstrating emergency service capabilities in authentic scenarios and advancing best practices for safeguarding licensed digital resources.

C Community
This drill serves as a compelling demonstration of how a national preservation community—encompassing 25 libraries, Karger Publishers, and the NDPC—can effectively mobilize collective expertise and resources to confront complex digital preservation challenges that exceed the capabilities of any individual institution.

H Horizon
In response to geopolitical and technological changes, this drill aims to establish a forward-looking framework to develop flexible and robust preservation systems that ensure continuous access to critical biomedical knowledge—maintaining a clear focus on future horizons.

6 Future Outlook

- **Enhance Systems:** Establish annual emergency response exercises as standard practice
- **Expand Coverage:** Extend the emergency drill model to additional disciplines and publishers
- **International Collaboration:** Share China's experience with global preservation communities

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NDPC | Medical Branch of NPDC | Karger Publisher | DRAA Alliance

Karger Electronic Journals → Full-text Search & Browse → Storage Services → Data Security Validation

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